



Data Structure

2024 Spring

Assignment #3: Dynamic Programming

Yunyong Ko

School of Computer Science and Engineering

Chung-Ang University

Assignment #3: Dynamic Programming

□ Overview

- Implement a dynamic-programming approach to the LCS problem
 - LCS: Longest common subsequence

□ Functions to implement

- Function for computing **the maximum length of a common subsequence** between two sequences and finding **the LCS itself**
- Function for printing the LCS and its length
- Function for finding the **most similar** sequence of a given sequence

Assignment #3: Dynamic Programming

□ Functions to implement

- Function for computing the maximum length of a common subsequence between two sequences and finding the LCS itself

- Please use the bottom-up dynamic programming approach (i.e., tabulation)
- **Input:** two sequences
- **Output:** a longest common subsequence between the two sequences and its length

- Function for printing the LCS and its length

LCS-LENGTH(X, Y)

```
1   $m = X.length$ 
2   $n = Y.length$ 
3  let  $b[1..m, 1..n]$  and  $c[0..m, 0..n]$  be new tables
4  for  $i = 1$  to  $m$ 
5       $c[i, 0] = 0$ 
6  for  $j = 0$  to  $n$ 
7       $c[0, j] = 0$ 
8  for  $i = 1$  to  $m$ 
9      for  $j = 1$  to  $n$ 
10         if  $x_i == y_j$ 
11              $c[i, j] = c[i - 1, j - 1] + 1$ 
12              $b[i, j] = \nwarrow$ 
13         elseif  $c[i - 1, j] \geq c[i, j - 1]$ 
14              $c[i, j] = c[i - 1, j]$ 
15              $b[i, j] = \uparrow$ 
16         else  $c[i, j] = c[i, j - 1]$ 
17              $b[i, j] = \leftarrow$ 
18  return  $c$  and  $b$ 
```

		j						
		0	1	2	3	4	5	6
i	y_j	B	D	C	A	B	A	
0	x_i	0	0	0	0	0	0	0
1	A	0	↑	↑	↑	←	←	←
2	B	0	1	←	←	1	2	←
3	C	0	↑	↑	2	←	2	↑
4	B	0	↑	↑	↑	2	3	←
5	D	0	↑	2	↑	2	↑	↑
6	A	0	↑	↑	2	3	↑	4
7	B	0	1	2	2	3	4	4

Assignment #3: Dynamic Programming

□ Functions to implement

- Function for finding the **most similar** sequence of a given sequence
 - **Input:** n sequences, target sequence index
 - **Output:** the most similar sequence to the target sequence and its length and index
 - The most similar sequence: a sequence has the maximum length of a common subsequence
 - When there are multiple sequences having the maximum length of a common subsequence, pick anyone

Target sequence

$X_1 = \text{ACCGGTC}$

$X_2 = \text{CACGGAT}$

$\vdots$

$X_n = \text{TCATGTC}$

Input



$X_k = \text{ACGCTCA}$

Output

Length=5 Index=k

Assignment #3: Input & Output

□ Input file

- The first line has an integer value which is the maximum length of a sequence
- From the the second line, each line represents a sequence of DNA

1st sequence ← 1 50
2 TTATACGTAGACACCAGATCCCAGGACATCTTATGATCATCTCATCCTGT
3 TGTCGTGCTCGCTAAAGCTATGAGCCCTCTCAAGCGGGTTTTGTAAAGC
4 CCGGAACGAGGAGTCGTTTCGATATGCATTAGTCTCACAGTGGCACCACC
5 GCGGGAGAAAAGTTTTTTTCGTAGGCACCTTGGCCGGCCCGCCGAAGATC
6 GCATCTGGATCCGTCGGTGGCCATCCATGAACTAAAGTAAGAGGTGACG
7 CTCCTGATCTCAGACACAGATCTAGTGGACAACATTTGCGGTTGCCTG
8 CAGTGACTAAGAAGGCTCGTTAGGTTGCTTCCATTGCGCTCCGATTCAAG
9 GAGATCGATAGAGCCGTTGTTAGAGATACTAGACCGCAATAGTCTAAC
10 TCACAGCACAAGGATTGTGTAGTACCACACACTCATTTCAACACTCG
k-th line represents (k-1)-th sequence ← 11 CTTTCCGCTTATTTGAACTCTGGGGTTTTCCATAGCAAATTGCCAGCCAT
12 GTCACATGCACTGTAAGATACCCGTAATCTTTGAAGCTGAGTGAATCC
13 CCCAGACAAGCTTCTATAAAGCTGTTATTACTTTTATGTGAACCTTTC
14 TAACATTGCCTTCTGTAGTGGTTCCAGATCCCGTCCCTCCTTTCAAGTGC
15 CAATAACGCCCGGGGATTGGAATCTACAATAACCTTGTAACTAGGCTCA
16 ACTCCCGCATATTGAAGGAAGATTGGATCTACTAATTTAGGTAAGCGCAG
17 AGACCGTCACCGTCACGCGTGTCTCATAAGCACGAATATTGAAATAAAA
18 ACTCAACCGCGACTTGGGAGTCCGTCGGTCGGCTGGTCTCTTGACAAGCC
19 CTCTGAAAGTAATCGCTGGGATTGATACGGACTCGACCACTCATGTTTG
20 GCCATTCCGGAGAATAGCGCCTAGCTGCAAGGCCAAACACCGACTGGAC
20th sequence ← 21 CATCCCCACCTCGCACCTCCCATAATGCTTGAAGTCCTTCATAGGGACCC

Example of Input.txt

Assignment #3: Input & Output

Output file

- The k-th line has the information of *the most similar sequence to the k-th sequence*
 - Format: <the index of the most similar sequence> <tab> <the length of the LCS> <tab> <the LCS>
 - i.e., (the number of lines in the input file – 1) == (the number of lines in the output file)

1st sequence ←

```
1 6 36 TTATCTAGACACAGATCAGGACACTTTGCTTCCCTG
2 17 33 CTCCGCGCTTGAGCCTCTCGGGGTTTTGAAAGC
3 7 33 CGGACAGAGTCGTTATTGCTTCATCTCCGTCAA
4 7 31 CGGAAAAGTCGTAGGCCCTTGCGCCGCAAG
5 14 33 CAATCGCCGGGGATTGAACTAAAGTAAGAGGTC
6 1 36 TTATCTAGACACAGATCAGGACACTTTGCTTCCCTG
7 3 33 CGGAAGAGGTCGTTGTTGCATTGCTCCATCAA
8 7 33 AGTGATAGAGGTGTTAGGTTGCCATGTCCTAA
9 3 33 CCGAAAGGAGTGTGTACACACCACATCACCACC
10 15 33 CTCCGCATTTGAAGGGTTTCTACAATTGAGCCA
11 9 33 TCACAGCACTGTAGTACCCAAATCTTCAAACC
12 7 33 CAGTCTAAAAGCTGTTATTCTTCATTGCCTTTC
13 14 33 TAACGCCCGGGGTTTCGATCTCTCCTTTCAGGTC
14 15 34 ATCCCCGGGGATTGATCTACTAATTTAGTAGCCA
15 14 34 CTCCCCGGGGATTGATCTACTAATTTAGTAGGCA
16 5 33 GCTCACCGTCCGGTGCCATCCATATAAATAAAA
17 2 33 TCCCGCACTTGAGCCTCTCGCGGTTTTGAAAGC
18 1 32 TTAAGTAACCAGATCGGACTCGACACTCATTG
19 5 33 GCATTCCGGGTGGCCACTGAACAAAAGAGGAC
20 11 33 TCACTGCACTCCCTAATCTTGAAGCTAGGGACC
```

20th sequence ←

No space at the end of each line

Example of output.txt

Assignment #3: Makefile

□ Writing a MakeFile

```
all: compile

compile: lcs.c
|   gcc lcs.c -o lcs

run: lcs
|   ./lcs input.txt output.txt

clean: lcs
|   rm lcs
```

```
run: lcs.py
|   python3 lcs.py input.txt output.txt
```

Assignment #3: Evaluation



- ☐ **Makefile**
- ☐ **Comments for all functions that you implemented**
- ☐ **Time complexity of the algorithms that you implemented**
- ☐ **Results on test input files**

Assignment #3: Submission Guideline

□ Submission

■ What: **Compressed file (Source code, input file, makefile)**

- Filename: **Assignment3_studentID**
- ***Compress the three files together**, **NOT** your workspace directory
 - Or name your workspace directory as **Assignment3_studentID**

■ Where: **Assignment board on e-class**

■ Deadline: **11:59 PM, May 17th (Friday), 2024**

- +1 day (75%), +2 days (50%), +3 days (**Not accepted**)

□ Q&A

■ Office hour: 3:00 PM, Tuesday

■ Contact Information

- Email: yyko@cau.ac.kr
- Tel: 02-820-5507

Q&A:
Assignment #3: Dynamic Programming